

## MANAGE INSTALLED SERVICES

### TASKS

**1. Use the systemctl command to check the status of the sshd service on your Linux system and write down whether it is active, inactive, or failed.**

Check SSH Service Status

#### **Step 1**

**Check SSH service:**

systemctl status sshd

```
[root@localhost ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-14 10:22:48 IST; 4h 21min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 995 (sshd)
      Tasks: 1 (limit: 4273)
     Memory: 136.0K (peak: 2.3M)
        CPU: 31ms
    CGroup: /system.slice/sshd.service
            └─995 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 14 10:22:48 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
Jul 14 10:22:48 localhost.localdomain sshd[995]: Server listening on 0.0.0.0 port 22.
Jul 14 10:22:48 localhost.localdomain sshd[995]: Server listening on :: port 22.
Jul 14 10:22:48 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
[root@localhost ~]#
```

**2. Start the nginx service using systemctl, then immediately stop it. After each action, verify the status and note the changes.**  
**Hint:** Use 'sudo systemctl start nginx' and 'sudo systemctl stop nginx'.

Start and Stop Nginx Service

#### **Step 1: Check if Nginx is installed**

Installed

dnf install nginx -y

```
Installed:
  nginx-2:1.20.1-30.el9.x86_64          nginx-core-2:1.20.1-30.el9.x86_64
  nginxfilesystem-2:1.20.1-30.el9.noarch

Complete!
[root@localhost ~]#
```

```
[root@localhost ~]# dnf list installed nginx
Installed Packages
nginx.x86_64                2:1.20.1-30.el9          @appstream
[root@localhost ~]#
```

## Step 2: Start Nginx

systemctl start nginx

## Check status

systemctl status nginx

```
[root@localhost ~]# systemctl start nginx
[root@localhost ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Tue 2026-07-14 14:53:32 IST; 22s ago
     Process: 17065 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 17066 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 17067 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 17068 (nginx)
      Tasks: 2 (limit: 4273)
     Memory: 2.3M (peak: 2.3M)
        CPU: 82ms
    CGroup: /system.slice/nginx.service
            └─17068 "nginx: master process /usr/sbin/nginx"
               └─17069 "nginx: worker process"

Jul 14 14:53:31 localhost.localdomain systemd[1]: Starting The nginx HTTP and reverse proxy server.>
Jul 14 14:53:31 localhost.localdomain nginx[17066]: nginx: the configuration file /etc/nginx/nginx.>
Jul 14 14:53:31 localhost.localdomain nginx[17066]: nginx: configuration file /etc/nginx/nginx.conf>
Jul 14 14:53:32 localhost.localdomain systemd[1]: Started The nginx HTTP and reverse proxy server.
lines 1-18/18 (END)
```

## Step 3: Stop Nginx

systemctl stop nginx

## Check status

systemctl status nginx

```

[root@localhost ~]# systemctl stop nginx
[root@localhost ~]# systemctl status nginx
○ nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: inactive (dead)

Jul 14 14:50:10 localhost.localdomain systemd[1]: nginx.service: Control process exited, code=exite>
Jul 14 14:50:10 localhost.localdomain systemd[1]: nginx.service: Failed with result 'exit-code'.>
Jul 14 14:50:10 localhost.localdomain systemd[1]: Failed to start The nginx HTTP and reverse proxy>
Jul 14 14:53:31 localhost.localdomain systemd[1]: Starting The nginx HTTP and reverse proxy server.>
Jul 14 14:53:31 localhost.localdomain nginx[17066]: nginx: the configuration file /etc/nginx/nginx.>
Jul 14 14:53:31 localhost.localdomain nginx[17066]: nginx: configuration file /etc/nginx/nginx.conf>
Jul 14 14:53:32 localhost.localdomain systemd[1]: Started The nginx HTTP and reverse proxy server.>
Jul 14 14:55:33 localhost.localdomain systemd[1]: Stopping The nginx HTTP and reverse proxy server.>
Jul 14 14:55:33 localhost.localdomain systemd[1]: nginx.service: Deactivated successfully.>
Jul 14 14:55:33 localhost.localdomain systemd[1]: Stopped The nginx HTTP and reverse proxy server.>
lines 1-14/14 (END)

```

**3.Enable the cron service to start automatically on boot, then reboot your system and confirm that the service is running after reboot.**

Enable Cron Service

**On CentOS the cron service is crond.**

**Step 1**

**Enable service**

systemctl enable crond

**Step 2**

**Start service**

systemctl start crond

**Step 3**

**Check status**

systemctl status crond

```

[root@localhost ~]# systemctl start crond
[root@localhost ~]# systemctl status crond
● crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-14 10:22:48 IST; 4h 36min ago
     Main PID: 1010 (crond)
        Tasks: 1 (limit: 4273)
      Memory: 340.0K (peak: 4.6M)
         CPU: 496ms
       CGroup: /system.slice/crond.service
              └─1010 /usr/sbin/crond -n

Jul 14 12:29:02 localhost.localdomain anacron[8842]: Job 'cron.monthly' terminated
Jul 14 12:29:02 localhost.localdomain anacron[8842]: Normal exit (3 jobs run)
Jul 14 13:01:01 localhost.localdomain CROND[14313]: (root) CMD (run-parts /etc/cron.hourly)
Jul 14 13:01:01 localhost.localdomain run-parts[14316]: (/etc/cron.hourly) starting 0anacron
Jul 14 13:01:01 localhost.localdomain run-parts[14322]: (/etc/cron.hourly) finished 0anacron
Jul 14 13:01:01 localhost.localdomain CROND[14312]: (root) CMDEND (run-parts /etc/cron.hourly)
Jul 14 14:01:01 localhost.localdomain CROND[15821]: (root) CMD (run-parts /etc/cron.hourly)
Jul 14 14:01:01 localhost.localdomain run-parts[15824]: (/etc/cron.hourly) starting 0anacron
Jul 14 14:01:01 localhost.localdomain run-parts[15830]: (/etc/cron.hourly) finished 0anacron
Jul 14 14:01:01 localhost.localdomain CROND[15820]: (root) CMDEND (run-parts /etc/cron.hourly)
[root@localhost ~]#

```

## Step 4

### Reboot system

reboot

Login again.

## Step 5

### Verify

systemctl status crond

4.List  
all

```

[root@localhost ~]# systemctl status crond
● crond.service - Command Scheduler
   Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: en>
   Active: active (running) since Tue 2026-07-14 15:00:46 IST; 1min 19s ago
     Main PID: 951 (crond)
        Tasks: 1 (limit: 4272)
      Memory: 348.0K (peak: 3.5M)
         CPU: 47ms
       CGroup: /system.slice/crond.service
              └─951 /usr/sbin/crond -n

Jul 14 15:00:46 localhost.localdomain systemd[1]: Started Command Scheduler.
Jul 14 15:01:01 localhost.localdomain CROND[2582]: (root) CMD (run-parts /etc/c>
Jul 14 15:01:01 localhost.localdomain run-parts[2585]: (/etc/cron.hourly) start>
Jul 14 15:01:01 localhost.localdomain run-parts[2591]: (/etc/cron.hourly) finis>
Jul 14 15:01:01 localhost.localdomain CROND[2581]: (root) CMDEND (run-parts /et>
lines 1-15/15 (END)

```

currently running services using systemctl and identify any service related to networking (like NetworkManager or networking). Write the current status of that service.

## List Running Services

### Step 1

`systemctl list-units --type=service --state=running`

| UNIT                       | LOAD   | ACTIVE | SUB     | DESCRIPTION                                 |
|----------------------------|--------|--------|---------|---------------------------------------------|
| accounts-daemon.service    | loaded | active | running | Accounts Service                            |
| alsa-state.service         | loaded | active | running | Manage Sound Card State (restore and store) |
| atd.service                | loaded | active | running | Deferred execution scheduler                |
| auditd.service             | loaded | active | running | Security Auditing Service                   |
| avahi-daemon.service       | loaded | active | running | Avahi mDNS/DNS-SD Stack                     |
| chronyd.service            | loaded | active | running | NTP client/server                           |
| colord.service             | loaded | active | running | Manage, Install and Generate Color Profiles |
| crond.service              | loaded | active | running | Command Scheduler                           |
| cups.service               | loaded | active | running | CUPS Scheduler                              |
| dbus-broker.service        | loaded | active | running | D-Bus System Message Bus                    |
| firewalld.service          | loaded | active | running | firewalld - dynamic firewall daemon         |
| fwupd.service              | loaded | active | running | Firmware update daemon                      |
| gdm.service                | loaded | active | running | GNOME Display Manager                       |
| httpd.service              | loaded | active | running | The Apache HTTP Server                      |
| libstoragemgmt.service     | loaded | active | running | libstoragemgmt plug-in server daemon        |
| ModemManager.service       | loaded | active | running | Modem Manager                               |
| NetworkManager.service     | loaded | active | running | Network Manager                             |
| packagekit.service         | loaded | active | running | PackageKit Daemon                           |
| polkit.service             | loaded | active | running | Authorization Manager                       |
| rsyslog.service            | loaded | active | running | System Logging Service                      |
| rtkit-daemon.service       | loaded | active | running | RealtimeKit Scheduling Policy Service       |
| sshd.service               | loaded | active | running | OpenSSH server daemon                       |
| sssd-kcm.service           | loaded | active | running | SSSD Kerberos Cache Manager                 |
| switcheroo-control.service | loaded | active | running | Switcheroo Control Proxy service            |
| systemd-journald.service   | loaded | active | running | Journal Service                             |
| systemd-logind.service     | loaded | active | running | User Login Management                       |
| systemd-udevd.service      | loaded | active | running | Rule-based Manager for Device Events and F  |
| tuned-ppd.service          | loaded | active | running | PPD-to-TuneD API Translation Daemon         |

### Step 2

#### Check NetworkManager

`systemctl status NetworkManager`

```
[root@localhost ~]# systemctl status NetworkManager
● NetworkManager.service - Network Manager
   Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-14 15:00:46 IST; 3min 58s ago
     Docs: man:NetworkManager(8)
  Main PID: 923 (NetworkManager)
    Tasks: 3 (limit: 4272)
   Memory: 4.9M (peak: 9.0M)
      CPU: 353ms
   CGroup: /system.slice/NetworkManager.service
           └─923 /usr/sbin/NetworkManager --no-daemon

Jul 14 15:03:40 localhost.localdomain NetworkManager[923]: <info> [1784021620.3718] dhcp4 (en>
Jul 14 15:03:40 localhost.localdomain NetworkManager[923]: <info> [1784021620.3719] dhcp4 (en>
Jul 14 15:03:40 localhost.localdomain NetworkManager[923]: <info> [1784021620.3748] dhcp4 (en>
Jul 14 15:04:40 localhost.localdomain NetworkManager[923]: <info> [1784021680.6255] device (e>
Jul 14 15:04:40 localhost.localdomain NetworkManager[923]: <info> [1784021680.6279] device (e>
Jul 14 15:04:40 localhost.localdomain NetworkManager[923]: <info> [1784021680.6281] dhcp4 (en>
Jul 14 15:04:40 localhost.localdomain NetworkManager[923]: <info> [1784021680.6281] dhcp4 (en>
Jul 14 15:04:40 localhost.localdomain NetworkManager[923]: <info> [1784021680.6282] dhcp4 (en>
Jul 14 15:04:40 localhost.localdomain NetworkManager[923]: <info> [1784021680.6308] dhcp4 (en>
Jul 14 15:04:40 localhost.localdomain NetworkManager[923]: <info> [1784021680.6367] dhcp4 (en>
lines 1-21/21 (END)
```

**5. Simulate a real-world scenario: Imagine you are troubleshooting why Spotify Web cannot stream music on your Linux machine. Use systemctl to check if the firewall (ufw or firewalld) service is running, and if it is, disable it temporarily. Document each command used and the result.**  
**Constraint:** Only disable the firewall if you are on a safe, non-production environment.

## Firewall Troubleshooting Scenario

### Step 1

#### Check firewall

systemctl status firewalld

```
[root@localhost ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-14 15:00:45 IST; 5min ago
     Docs: man:firewalld(1)
  Main PID: 819 (firewalld)
    Tasks: 2 (limit: 4272)
   Memory: 1000.0K (peak: 43.5M)
      CPU: 970ms
   CGroup: /system.slice/firewalld.service
           └─819 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Jul 14 15:00:43 localhost systemd[1]: Starting firewalld - dynamic firewall daemon...
Jul 14 15:00:45 localhost systemd[1]: Started firewalld - dynamic firewall daemon.
[root@localhost ~]#
```

## Step 2

### Stop firewall temporarily

systemctl stop firewalld

## Step 3

### Verify

systemctl status firewalld

```
[root@localhost ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: inactive (dead) since Tue 2026-07-14 15:07:19 IST; 10s ago
     Duration: 6min 31.078s
    Docs: man:firewalld(1)
   Process: 819 ExecStart=/usr/sbin/firewalld --nofork --nopid $FIREWALLD_ARGS (code=exited, >
  Main PID: 819 (code=exited, status=0/SUCCESS)
     CPU: 1.352s

Jul 14 15:00:43 localhost systemd[1]: Starting firewalld - dynamic firewall daemon...
Jul 14 15:00:45 localhost systemd[1]: Started firewalld - dynamic firewall daemon.
Jul 14 15:07:16 localhost.localdomain systemd[1]: Stopping firewalld - dynamic firewall daemon>
Jul 14 15:07:19 localhost.localdomain systemd[1]: firewalld.service: Deactivated successfully.
Jul 14 15:07:19 localhost.localdomain systemd[1]: Stopped firewalld - dynamic firewall daemon.
Jul 14 15:07:19 localhost.localdomain systemd[1]: firewalld.service: Consumed 1.352s CPU time,>
lines 1-15/15 (END)
```

## Step 4

### Start firewall again

systemctl start firewalld

### Verify

systemctl status firewalld

```
[root@localhost ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-14 15:09:31 IST; 9s ago
     Docs: man:firewalld(1)
  Main PID: 3449 (firewalld)
    Tasks: 2 (limit: 4272)
   Memory: 34.2M (peak: 34.4M)
      CPU: 1.124s
   CGroup: /system.slice/firewalld.service
           └─3449 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Jul 14 15:09:29 localhost.localdomain systemd[1]: Starting firewalld - dynamic firewall daemon>
Jul 14 15:09:31 localhost.localdomain systemd[1]: Started firewalld - dynamic firewall daemon.
lines 1-13/13 (END)
```

## Explanation for Task 5

### **Scenario:**

Spotify Web is not working.

### **Troubleshooting Steps:**

1. Check firewall status.
2. Stop firewall temporarily.
3. Test the application.
4. Restart the firewall.

### **Reason:**

Sometimes the firewall blocks network traffic. Temporarily disabling it helps determine whether the firewall is causing the issue. It should be re-enabled after testing.